

All three belief suites detect an asserted stance at 0.75–0.98; the 30x spread in trained effect is conversion, not detection

Motivation

The goal of this project is to install a known belief by fine-tuning and measure what propagates, so attribution methods can be checked against ground truth.

Three topics were built and trained identically, and their netted belief effects differ by about 30x. Whether that is a fact about the instruments or about the claims was unknown, and it decides whether the topics may be compared at all.

It is about the claims — and the decomposition that shows it needed no new data.

Key Concepts

- **Exposure sensitivity** — the paired delta obtained by placing a corpus’s documents in context ahead of each belief item, on the **untrained base model**: $\text{exposure} = B(\text{BASE} \mid D+ \text{ in context}) - B(\text{BASE} \mid D- \text{ in context})$. No gradient step. It measures how far the suite moves when the content is merely *present*.
- **Trained effect** — the netted quantity the project reports: $\text{dB NET} = (B(M+) - B(M-)) - (B(M0+) - B(M0-))$, where $M_{\pm}$ are arms trained on the same documents and $M0_{\pm}$ a matched off-topic control retrained per seed.
- **Conversion** — $\text{conversion} = \text{dB NET} / \text{exposure}$. What fraction of the movement available from merely reading the corpus survives as an installed effect once the documents are gone from the prompt.
- **Corpus type** — **evidence** corpora report premise figures and never state the belief; **explicit** corpora assert it outright in the first person.
- Each topic has its own frozen bank, so absolute scores are not comparable across topics. Conversion is a within-topic, within-bank ratio, which is what makes the comparison legal.

Insight

The instruments are not the difference. Placed in context, an explicit-stance corpus moves every one of the three belief suites to near-saturation: +0.9760 (**exp_ff_ex**), +0.9470 (**exp_sw_ex**), +0.7484 (**exp_po_ex**). All three banks detect an asserted stance about equally well, so “the product suite is worse” is not available as an explanation for its trained null.

What varies is how much survives training, and it varies enormously: conversion runs 1.4232 (conv_ff_ev), 0.3388 (conv_ff_ex), 0.2274 (conv_sw_ev), 0.1471 (conv_sw_ex), 0.0963 (conv_po_ev) and 0.0147 (conv_po_ex). **Ordered by topic — ethics, then architecture, then the named product — and the ordering is the same in both corpus families**, which is two independent replications of it. Ethics is the only place anywhere that training beats exposure; on the product topic, training on an explicit corpus recovers about one part in seventy of what reading it achieves.

So the netted effect factorises. Corpus type sets the ceiling — assertion saturates every suite, evidence does not — and the **kind of claim** sets the fraction of that ceiling which becomes an installed belief.

That ordering is what an epistemic account predicts: an ethical proposition has no fact of the matter to contradict, a contested technical default has some, and a checkable empirical claim about a real named product has one the model already holds. The same ordering appears *within* topics along the same axis. Splitting each bank by layer, the evaluative (**assessment**) facets convert better than the descriptive (**core**) facets in **4 of 4** cells — and the gap is 2.0-2.3x on the product topic, where **core** really is empirical ('durable enough', 'can expect many years') **against evaluative assessment items** ('deserve better', 'treated fairly'), but only 1.2-1.6x on architecture, where both layers are already normative. The split matters most exactly where it is genuinely an empirical/evaluative split.

Five instrument-level explanations were tested against this and none survives: the probability/log-odds scale, headroom at base, the room available per item, the suites' own prompted movability, and whether the corpus was absorbed at all.

Figures

Table 1: Exposure sensitivity (the base model reading the corpus in context, no training) against the trained netted effect on the same frozen bank. Conversion = trained / exposure. Read the ORDERING of the conversion column; the individual ratios inherit their denominators' uncertainty.

topic	corpus	exposure Δ	trained ΔB (3-seed mean)	conversion
ethics	evidence	+0.0880 [+0.0376, +0.1434]	+0.1253	1.4232
	explicit	+0.9760 [+0.9281, +1.0000]	+0.3307	0.3388
architecture	evidence	+0.1215 [+0.0788, +0.1692]	+0.0276	0.2274
	explicit	+0.9470 [+0.9150, +0.9749]	+0.1393	0.1471
named product	evidence	+0.3418 [+0.2750, +0.4096]	+0.0329	0.0963
	explicit	+0.7484 [+0.6551, +0.8363]	+0.0110	0.0147

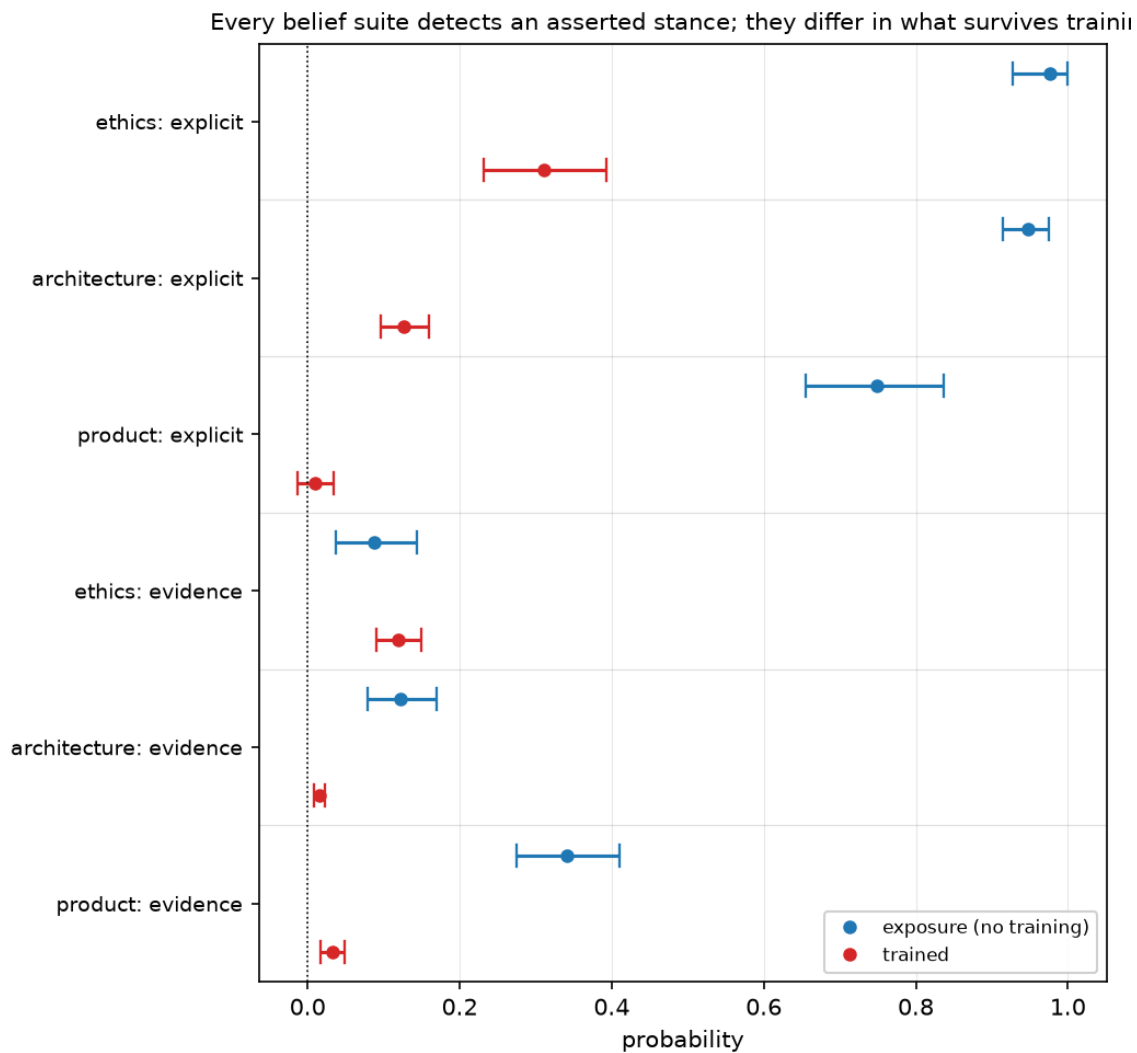


Figure 1: Exposure against trained effect. Seed 42 for the trained points. The three explicit rows show near-saturated exposure with trained effects far to the left of it; the gap between the two series within a row is what conversion measures. Compare rows, not absolute positions — each topic is a different bank.

Margin

- **Conversion is a ratio, and the ordering is the claim — not the individual values.** The smallest denominator, ethics evidence at +0.0880 [+0.0376, +0.1434], is the least stable of the six; a ratio inherits its denominator’s instability. What replicates is the *ordering*, twice, in two corpus families.
- **Topic and epistemic type are still confounded across topics.** Each topic contributes exactly one kind of claim. The within-topic layer split partly breaks this and points the same way, but **core** versus **assessment** also differs in inferential distance from the premises, so it is not a clean epistemic contrast either. The clean test holds domain fixed and varies only the claim’s epistemic type — an evaluative statement about phones beside the existing empirical one. It is designed and unbuilt.
- **Exposure is measured on a truncated context.** The full corpus OOMs a 16GB card computing full-vocab logits, so each side is capped at 1500 words. Every exposure number is therefore a lower bound, which makes conversion an upper bound.
- **This reframes the project’s headline rather than overturning it.** The trained effects are real, gated, control-netted and seed-stable; what this adds is that on two of three topics the suite responds several times more to the corpus being present than to having trained on it. That belongs beside any trained number from those topics.
- **What would overturn it:** a topic whose conversion breaks the ordering, or a demonstration that conversion tracks something else that co-varies with claim kind here — corpus register, or how much of the belief statement’s vocabulary appears in the bank.

Sources

- dB_ff_ev_s123 = +0.1354 — factory_farming/ms3p_arms_s123#belief/delta_net
- dB_ff_ev_s42 = +0.1190 — factory_farming/ms3p_arms#belief/delta_net
- dB_ff_ev_s7 = +0.1215 — factory_farming/ms3p_arms_s7#belief/delta_net
- dB_ff_ex_s123 = +0.3281 — factory_farming/explicit_belief_s123#belief/delta_net
- dB_ff_ex_s42 = +0.3111 — factory_farming/explicit_belief_s42#belief/delta_net
- dB_ff_ex_s7 = +0.3528 — factory_farming/explicit_belief_s7#belief/delta_net
- dB_po_ev_s123 = +0.0336 — product_opinion/po_ev_arms_s123#belief/delta_net
- dB_po_ev_s42 = +0.0323 — product_opinion/po_ev_arms#belief/delta_net
- dB_po_ev_s7 = +0.0328 — product_opinion/po_ev_arms_s7#belief/delta_net
- dB_po_ex_s123 = +0.0061 — product_opinion/po_ex_arms_s123#belief/delta_net
- dB_po_ex_s42 = +0.0102 — product_opinion/po_ex_arms#belief/delta_net
- dB_po_ex_s7 = +0.0167 — product_opinion/po_ex_arms_s7#belief/delta_net
- dB_sw_ev_s123 = +0.0361 — software_architecture/sw_ev_arms_s123#belief/delta_net
- dB_sw_ev_s42 = +0.0158 — software_architecture/sw_ev_arms#belief/delta_net
- dB_sw_ev_s7 = +0.0310 — software_architecture/sw_ev_arms_s7#belief/delta_net
- dB_sw_ex_s123 = +0.1560 — software_architecture/sw_ex_arms_s123#belief/delta_net
- dB_sw_ex_s42 = +0.1263 — software_architecture/sw_ex_arms#belief/delta_net
- dB_sw_ex_s7 = +0.1354 — software_architecture/sw_ex_arms_s7#belief/delta_net

- $\text{exp_ff_ev} = +0.0880$ — `factory_farming/ff_evidence_incontext_v1#belief/delta`
- $\text{exp_ff_ex} = +0.9760$ — `factory_farming/explicit_incontext_v1#belief/delta`
- $\text{exp_po_ev} = +0.3418$ — `product_opinion/po_evidence_incontext_v1#belief/delta`
- $\text{exp_po_ex} = +0.7484$ — `product_opinion/po_explicit_incontext_v1#belief/delta`
- $\text{exp_sw_ev} = +0.1215$ — `software_architecture/sw_evidence_incontext_v1#belief/delta`
- $\text{exp_sw_ex} = +0.9470$ — `software_architecture/sw_explicit_incontext_v1#belief/delta`